

CHAPTER 17 Part 1 — Understanding ANN from Absolute Basics

Chapter 17 — Artificial Neural Network (ANN) Part 1 — Understanding ANN from Absolute Basics Before Starting Congratulations. Until now our project has been centered around Support Vector Machine. Question. Did our project use only SVM? No. We also implemented an

Artificial Neural Network

or ANN. Question. Why? Because we wanted to compare two different Machine Learning approaches. One classical. One deep learning. That comparison made our project much stronger.

What you will know after finishing this chapter

After this chapter, you should be able to answer [OK] What is an Artificial Neural Network? [OK] Why is it called a Neural Network? [OK] What is a neuron? [OK] What are layers? [OK] How does information flow? [OK] Why did we compare ANN with SVM?

Difficulty

Basics 5/5

Theory 5/5

Coding 1/5

Viva Importance 5/5

Let's Begin

What is an Artificial Neural Network? Simple definition. An ANN is a Machine Learning model inspired by the way the human brain processes information. Notice one important word. Inspired. Not identical. It is only loosely inspired by biological neurons.

PROFESSOR ASKS

What is an Artificial Neural Network?

EXCELLENT ANSWER

An Artificial Neural Network is a machine learning model inspired by the structure of biological neural networks. It learns complex relationships between inputs and outputs by adjusting internal weights during training. Excellent. Why Is It Called a Neural Network? Question. What is the brain made of? Neurons. Billions of them. Each neuron receives information, processes it, and passes it to other neurons. ANN tries to imitate that idea. Instead of biological neurons, it uses mathematical neurons.

Real Life Analogy

Imagine a company. **Customer** — Receptionist. **Manager** — Director. Decision. Information flows through different people. Exactly how ANN works. What Is an Artificial Neuron? Question. What is the smallest unit of an ANN? Answer. A neuron. Think of one neuron as a very small decision maker. It receives inputs, performs a calculation, and produces one output.

In Our Project

Question. What are the inputs? Our 17 features. Example Peak RMS Mean. DC Level These enter the first layer of the ANN.

PROFESSOR ASKS

What is a neuron?

EXCELLENT ANSWER

A neuron is the basic computational unit of an artificial neural network. It receives inputs, applies learned weights and an activation function, and produces an output.

Information Flow

Suppose our feature vector is [Peak, RMS, Mean, .. DC Level] Question. Where does it go? Into the Input Layer.

Input Layer

The input layer does almost nothing. It simply receives the features and passes them forward. Question. How many inputs did our ANN receive? Exactly Because our project contains 17 features.

PROFESSOR ASKS
Why does the input layer have 17 neurons?
EXCELLENT ANSWER

The input layer contains 17 neurons because each neuron receives one feature from the 17-dimensional feature vector extracted from the waveform.

Hidden Layer

Now the interesting part. After the input layer, the information moves to one or more Hidden Layers. Question. Why are they called Hidden? Because we cannot directly observe what they learn. The computations happen inside the network. What Happens Inside? Every neuron receives information from the previous layer. Each connection has a weight. Question. What is a weight? Think of it as the importance given to a particular input. **Important feature** — Higher influence. **Less important feature** — Lower influence.

Real Life Analogy

Imagine choosing a cricket player. Question. Which matters more? Batting or shoe size? Obviously batting. So batting gets more importance. Weights do exactly that.

PROFESSOR ASKS
What are weights?
EXCELLENT ANSWER

Weights are trainable numerical values that determine how strongly each input influences the output of a neuron.

Output Layer

Finally, the information reaches the Output Layer. Question. What does it produce? The predicted fault class. Example RC_Normal or RC_Open or RL_Normal Exactly like our SVM.

Information Flow Summary

The complete flow looks like

17 Features

Input Layer

Hidden Layer(s)

Output Layer

Fault Prediction

Simple. Why Use Hidden Layers? Question. Why not connect the input directly to the output? Because many relationships are too complex. Hidden layers allow the network to learn complicated nonlinear patterns. ANN vs SVM At this stage, don't compare performance. Just understand the philosophy. SVM tries to find the best decision boundary. ANN tries to learn a complex mapping from inputs to outputs. Different ideas. Same goal. Why Did We Build an ANN? Question. Did we expect ANN to automatically beat SVM? No. Our goal was comparison. We wanted to answer a scientific question. Which model works better for our analog fault diagnosis dataset? That is much stronger than using only one algorithm.

PROFESSOR ASKS

Why did you implement both SVM and ANN?

EXCELLENT ANSWER

Implementing both SVM and ANN allowed us to compare two fundamentally different machine learning approaches on the same dataset and identify the model that was better suited for analog fault diagnosis. Excellent.

Connection to Our Project

Our ANN pipeline looks like this.

17 Features

Input Layer

Hidden Layer(s)

Output Layer

Predicted Fault

Notice the feature extraction remains exactly the same. Only the classifier changes.

COMMON MISTAKES

MISTAKE: [X] ANN learns the circuit. Wrong. It learns the relationship between features and labels.

MISTAKE: [X] Hidden layers store the dataset. Wrong. They learn internal representations.

MISTAKE: [X] More hidden layers always mean better performance. Wrong. Too many layers can cause overfitting, especially on small datasets.

REVISION NOTES

Remember
the ANN structure.

17 Features — Input Layer.

Weights — Learn Importance.

Hidden Layers — Learn Patterns.

Output Layer — Predict Fault.

MEMORY TRICK

Remember this sentence. "Input receives, hidden learns, output predicts." Three phrases. Entire ANN architecture.

KNOWLEDGE CHECK

Level 1

- What is an Artificial Neural Network?
- Why is it called a neural network?
- What is a neuron?
- What is the input layer?
- What is the output layer?

Level 2

- Why does our ANN have 17 input neurons?
- What are weights?
- Why are hidden layers needed?
- How is ANN different from SVM?
- Why did we implement ANN in our project?

Professor Level

- Explain the complete ANN architecture used for analog fault diagnosis.
- Why are neural networks considered nonlinear models?
- How do weights influence the learning process?
- Why is comparing ANN with SVM scientifically valuable?
- What remains unchanged when replacing SVM with ANN in your project pipeline?
- End of Chapter 17 (Part 1)
- Congratulations.
- You now understand the basic architecture and philosophy of Artificial Neural Networks.
- Notice something important:
 - The dataset, feature extraction, and preprocessing pipeline remain exactly the same as in the SVM implementation. Only the classifier changes—from an SVM that learns an optimal hyperplane to an ANN that learns through interconnected layers of neurons.

Author's Note The next continuation will explain the actual ANN implementation used in your project, including: Input layer Hidden layers Output layer Activation functions (ReLU, Softmax, etc.) Forward propagation Loss function Optimizer (such as Adam) Epochs Batch size Why each of these choices was made in your implementation After that, we'll perform a complete SVM vs ANN comparison specifically for your analog fault diagnosis project.

CHAPTER 17 Part 2 — Our ANN Implementation

Chapter 17 — Artificial Neural Network (ANN) Part 2 — Our ANN Implementation (Line-by-Line Explanation) Before Starting

In Part 1

we understood what an Artificial Neural Network is. Now we'll study how our ANN was actually built. This chapter will explain every important part of the ANN code using the same style we used for SVM. Where Are We? Our pipeline is still the same. **LTspice** — CSV Files.

17 Features

Train-Test Split

StandardScaler — ANN ← We are here. Notice only the classifier has changed. Everything else remains the same.

Step 1 — Importing ANN Libraries

Before building the network, our code imports the required libraries. Conceptually it looks like

```
import tensorflow
```

```
from tensorflow.keras.models import Sequential
```

```
from tensorflow.keras.layers import Dense
```

Question. Why do we need these libraries? Because they provide ready-made tools for building Artificial Neural Networks.

PROFESSOR ASKS

Why did you use TensorFlow/Keras?

EXCELLENT ANSWER

TensorFlow and Keras provide efficient implementations of neural network layers, training algorithms, and optimization techniques, allowing us to build and train ANN models efficiently.

Step 2 — Creating the Model

Now our code creates an empty ANN. Conceptually model = Sequential() Question. What does Sequential mean? It means the layers are connected one after another. Like a straight pipeline. **Input** — Hidden. Output No branches. No shortcuts. Simple.

PROFESSOR ASKS

What is a Sequential model?

EXCELLENT ANSWER

A Sequential model is a neural network in which layers are arranged in a simple linear sequence, with the output of one layer becoming the input to the next.

Step 3 — Input Layer

Now we add the first layer. Question. How many neurons? Exactly Why? Because our feature vector contains 17 features. One neuron receives one feature. Simple. Question. Does the input layer learn anything? No. It simply passes the information forward.

Step 4 — Hidden Layer

Now the real learning begins. The hidden layer receives the 17 inputs and starts learning relationships. Every neuron performs three steps.

Receive Inputs

Multiply by Weights

Produce Output

This happens for every neuron. Question. What is being learned? The weights. Not the inputs. Not the labels. The weights.

Why Hidden Layers Matter

Imagine trying to recognize a face. Question. Do you look at only the eyes? No. You combine eyes, nose, mouth, hair, shape, everything. The hidden layer does exactly that. It combines multiple features to discover patterns.

PROFESSOR ASKS
What is learned during ANN training?
EXCELLENT ANSWER

During training, the ANN learns the optimal weights connecting neurons so that it can accurately map input feature vectors to the correct fault classes.

Step 5 — Activation Function

Now one of the most important concepts. Question. Suppose every neuron only performed simple addition. Would the ANN be powerful? No. Need something nonlinear. That is the Activation Function. ReLU The most common activation function used in hidden layers is ReLU. ReLU stands for Rectified Linear Unit. Question. What does ReLU do? Simple. **Negative** — 0. Positive

Remain Positive

That's all. Very simple.

Real Life Analogy

Imagine a security guard. Question. Negative visitors? Not allowed. Positive visitors? Allowed inside. ReLU works almost like that.

PROFESSOR ASKS
Why is ReLU used?
EXCELLENT ANSWER

ReLU introduces nonlinearity into the network while remaining computationally efficient. It allows the ANN to learn complex relationships between the input features and output classes.

Step 6 — Output Layer

Finally the information reaches the output layer. Question. How many neurons does the output layer contain? Answer. One neuron for each fault class. Suppose our dataset contains 10 classes. **Output layer** — 10 neurons. Each neuron represents one possible fault. Question. Which neuron wins? The neuron with the highest probability. That becomes the prediction.

PROFESSOR ASKS
Why does the output layer have one neuron per class?

EXCELLENT ANSWER

Each output neuron represents one possible fault class. The neuron with the highest output value corresponds to the predicted class.

Step 7 — Softmax

Question. Suppose the output layer produces these numbers. **RC_Normal** — 0.82. **RC_Open** — 0.09. **RL_Normal** — 0.05. Question. What do these numbers mean? They are probabilities. How are they produced? Using Softmax. Question. What does Softmax do? It converts raw outputs into probabilities whose total equals

PROFESSOR ASKS

Why Softmax?

EXCELLENT ANSWER

Softmax converts the output values into probabilities for all classes, allowing the network to predict the most likely fault class.

Step 8 — Loss Function

Question. How does the ANN know it made a mistake? Need a score. That score is called the Loss Function. Suppose the correct answer is RC_Open The network predicts RC_Normal Question. Good prediction? No. Loss becomes large. Question. Suppose prediction is correct. Loss becomes small. Simple.

PROFESSOR ASKS

What is a Loss Function?

EXCELLENT ANSWER

The loss function measures the difference between the predicted outputs and the actual labels. Training aims to minimize this loss.

Step 9 — Optimizer

Question. Suppose Loss is high. Question. How does the ANN improve itself? Using an Optimizer. In our project, the optimizer was Adam. Adam adjusts the weights to reduce the loss. Think of a student. Question. Gets many questions wrong. What happens? Studies. Improves. Scores better. Adam does the same thing for the network.

PROFESSOR ASKS

What is the role of Adam?

EXCELLENT ANSWER

Adam is an optimization algorithm that updates the neural network weights to minimize the loss function during training.

Step 10 — Epochs

Question. Does the ANN learn after seeing the dataset only once? Usually No. Need multiple passes. Each complete pass through the training dataset is called an Epoch. Example

Epoch 1

Better Model

PROFESSOR ASKS

What is an Epoch?

EXCELLENT ANSWER

An epoch is one complete pass of the entire training dataset through the neural network during training.

Step 11 — Batch Size

Question. Suppose the dataset contains 5000 samples. Question. Should the ANN process all 5000 at once? Usually No. Instead it processes small groups called Batches. The number of samples in each group is called the Batch Size. Complete ANN Pipeline Everything fits together.

17 Features

Input Layer

Hidden Layer

ReLU

Output Layer

Softmax — Loss.

Adam Optimizer

Weight Update

Repeat for Many Epochs — Final ANN. This is exactly how our ANN learns.

Connection to Our Project

Notice something interesting. The ANN received the exact same 17 features used by SVM. Only the learning algorithm changed. This makes our comparison fair. Both models used the same dataset, the same preprocessing, and the same features. Only the classifier was different.

COMMON MISTAKES

MISTAKE: [X] Epoch means one sample. Wrong.

MISTAKE: [X] Weights are fixed. Wrong. They change during training.

MISTAKE: [X] Softmax is used inside hidden layers. Wrong. It is normally used in the output layer for multiclass classification.

MISTAKE: [X] Adam is a neural network. Wrong. Adam is the optimizer.

REVISION NOTES

Remember the ANN workflow.

- Sequential** — Arrange Layers.
- Hidden Layers** — Learn Weights.
- ReLU** — Add Nonlinearity.
- Softmax** — Output Probabilities.
- Loss** — Measure Error.
- Adam** — Reduce Error.
- Epoch** — One Complete Pass.
- Batch** — Small Training Group.

MEMORY TRICK

Remember this sentence. "Features enter, weights learn, ReLU activates, Softmax predicts, Adam improves." One sentence. Entire ANN pipeline.

KNOWLEDGE CHECK

Level 1

- What is a Sequential model?
- Why does the input layer have 17 neurons?
- What does ReLU do?
- What is Softmax?
- What is an Epoch?

Level 2

- Why are activation functions necessary?
- What is the role of the optimizer?
- Why is Adam widely used?
- What is the difference between an Epoch and a Batch?
- Why did both ANN and SVM use the same feature vector?

Professor Level

- Explain the complete ANN implementation used in your project.
- Why is ReLU used in hidden layers and Softmax in the output layer?
- How does the optimizer reduce the loss?
- Why is keeping the same feature extraction pipeline important when comparing ANN and SVM?
- Describe the complete information flow from the input layer to the final prediction.
- End of Chapter 17 (Part 2)
- Congratulations.
- You now understand the complete ANN implementation, from building the network to training it and generating predictions.

Author's Note The next continuation is one of the most valuable chapters for interviews and vivas:

Chapter 18 — SVM vs ANN: Complete Comparison We'll compare them in every aspect: Working principle Training speed Dataset size requirements Overfitting Accuracy Interpretability Computational cost Why SVM performed better (or was chosen) for your project Which one would be better if the

dataset became much larger Interview questions comparing SVM and ANN This comparison is one of the topics interviewers frequently ask because it tests whether you understand why you selected a particular model, not just how to use it.

CHAPTER 18 Part 1 — SVM vs ANN: Complete Comparison

Chapter 18 — SVM vs ANN Part 1 — Complete Comparison (Which One Was Better for Our Project?) Before Starting Congratulations. Now you know both Machine Learning models used in our project. ✓ SVM ✓ ANN Question. Which one is better? Most beginners ask the wrong question. The correct question is Which model is better for this particular problem? There is no algorithm that is always the best. The answer depends on the dataset. Why Did We Compare Them? Question. Could we have used only SVM? Yes. Question. Then why build an ANN? Because we wanted to answer a scientific question. Which classifier is more suitable for analog circuit fault diagnosis? That made our project more convincing.

PROFESSOR ASKS

Why did you compare SVM and ANN?

EXCELLENT ANSWER

We compared SVM and ANN because they represent two fundamentally different machine learning approaches. Evaluating both models on the same dataset allowed us to determine which was more suitable for analog fault diagnosis. Excellent. Philosophy Let's first understand how they think. SVM SVM says Find the best decision boundary. It tries to separate different classes using the optimal hyperplane. ANN ANN says Learn the relationship between inputs and outputs by adjusting millions (or thousands) of weights. Different idea. Same goal.

Finds optimal hyperplane	Learns weights
Geometry-based	Brain-inspired
Support vectors	Neurons

Question. Suppose we have only 500 samples. Question. Which model usually performs better? Usually SVM. Question. Suppose we have 5 million samples. Question. Now? ANN usually becomes more powerful.

PROFESSOR ASKS

Which algorithm is better for small datasets?

EXCELLENT ANSWER

SVM generally performs better on small and medium-sized datasets because it can construct effective decision boundaries without requiring massive amounts of training data. Why? ANN has many weights. Many parameters. Needs lots of data. Otherwise it may overfit. SVM needs much less data.

Our Project

Question. Did our project have millions of samples? No. Our dataset was moderate in size. That already favored SVM. Comparison 3 Training Speed Question. Which trains faster? Usually SVM for small datasets. ANN needs many epochs. Repeated weight updates. Longer training. Question. If the dataset becomes extremely large? Now ANN can become more efficient using GPU acceleration. Comparison 4 Feature Engineering This is important. Question. Which model depends more on good features? SVM. Very much. Remember our 17 features. Peak. RMS. Energy. DC Level. Everything mattered. Question. Can ANN learn some useful internal representations? Yes. Hidden layers help discover complex patterns. Still, good features always help. Comparison 5 Interpretability Question. Which model is easier to explain? SVM. Why? Because we can explain

Support Vectors,

Margins, Hyperplanes. ANN contains many weights. Much harder to interpret.

PROFESSOR ASKS

Which model is more interpretable?

EXCELLENT ANSWER

SVM is generally more interpretable because its decision boundary and support vectors can be analyzed more easily than the large number of learned weights in an ANN. Comparison 6 Overfitting Question. Which model overfits more easily? Usually ANN. Especially when the dataset is small. Question. Can SVM also overfit? Yes. Especially if Gamma is too large or C is not tuned. Remember GridSearchCV. Comparison 7 Computational Cost Training ANN requires repeated forward propagation and backpropagation. Many epochs. Many weight updates. SVM solves an optimization problem once during training. For small datasets, this is usually faster. Comparison 8 Hyperparameters SVM needs Kernel C Gamma ANN needs many more.

Hidden Layers

Neurons

Learning Rate

Batch Size

Epochs Optimizer

Activation Functions

Dropout. Question. Which is easier to tune? Usually SVM.

Small Dataset	5/5	3/5
Large Dataset	3/5	5/5
Training Speed	Faster (small data)	Slower
Interpretability	High	Low
Hyperparameters	Few	Many
Overfitting Risk	Lower	Higher
GPU Benefit	Limited	Excellent
Feature Engineering	Very Important	Important

Now the important question. Question. Which model was more suitable for our project? Answer. SVM. Question. Why? Several reasons.

Reason 1

Dataset was moderate in size. SVM works extremely well on small and medium datasets.

Reason 2

Our features were already carefully engineered. Remember our 17 features. Peak. RMS. Energy. DC Level. Since the features already contained meaningful information, SVM could classify them very effectively.

Reason 3

The decision boundaries were complex, but not so complex that deep learning became necessary. The RBF kernel handled the nonlinear relationships very well.

Reason 4

SVM trained faster, was easier to optimize, and was easier to explain during the project.

PROFESSOR ASKS

Why was SVM selected as the primary model instead of ANN?

EXCELLENT ANSWER

The dataset size was moderate, the engineered features already captured meaningful circuit behavior, and the RBF-kernel SVM provided strong nonlinear classification with lower complexity and easier optimization than the ANN. Therefore, SVM was better suited for this project. This is an excellent placement and viva answer.

If The Dataset Became Huge

Question. Suppose instead of 5,000 samples, we had 5 million waveforms. Question. Would SVM still be the obvious choice? Maybe not. Now ANN could become much stronger because deep learning benefits from very large datasets.

Engineering Lesson

Notice something important. We did NOT choose our algorithm because it was popular. We chose it because it matched our problem. That is good engineering.

Connection to Our Project

Our comparison looked like this.

Same Dataset

Same Features

Same Train-Test Split

SVM — Results. AND

Same Dataset

Same Features

Same Train-Test Split

ANN — Results. Only the classifier changed. Everything else remained identical. That made the comparison fair.

COMMON MISTAKES

MISTAKE: [X] ANN is always better than SVM. Wrong.

MISTAKE: [X] Deep Learning should always replace classical ML. Wrong.

MISTAKE: [X] SVM works only for linear data. Wrong. The RBF kernel handles nonlinear data.

MISTAKE: [X] More complex model always means better accuracy. Wrong. Model choice depends on the dataset and the problem.

REVISION NOTES

Remember
the key differences.
SVM — Hyperplane.
ANN — Weights.
Small Dataset — SVM.
Huge Dataset — ANN.
Interpretability — SVM.
Complex Learning — ANN.

MEMORY TRICK

*Remember this sentence. "SVM is smarter with small data. ANN becomes stronger with big data."
One sentence. Entire comparison.*

KNOWLEDGE CHECK

Level 1

- Why did we compare SVM and ANN?
- Which model generally performs better on small datasets?
- Which model usually requires more hyperparameters?
- Which model is easier to interpret?
- Why was SVM chosen for our project?

Level 2

- Explain the main philosophical difference between SVM and ANN.
- Why does ANN generally require more training data?
- Why were our engineered features particularly suitable for SVM?
- How does dataset size influence model selection?
- Why is a fair comparison important in machine learning experiments?

Professor Level

- Compare SVM and ANN for analog fault diagnosis.
- Why did the RBF-kernel SVM outperform or become the preferred model in your project?
- Under what conditions would ANN become a better choice?
- Explain why algorithm selection should depend on the dataset rather than popularity.
- Describe the scientific methodology you used to compare both classifiers.
- End of Chapter 18 (Part 1)
- Congratulations.
- You now understand not just how SVM and ANN work, but why SVM was the better engineering choice for your specific analog fault diagnosis problem.
- This ability to justify why a model was selected is one of the strongest skills you can demonstrate in an interview or project viva.

Author's Note The next continuation will move beyond machine learning and into software engineering. We'll begin Chapter 19 — Dashboard Architecture, where we'll explain: Why the dashboard was built. Overall software architecture. Every button and UI element. Backend workflow. How the uploaded waveform moves through feature extraction, scaling, model loading, prediction, and visualization. How to explain the dashboard during a live project demonstration. This will complete the transition from machine learning model to a usable engineering application.